COMPUTING 2 - PARALLEL COMPUTING

Chapter 1 Mock Test

Performance Models and Parallel Design

Suggested time	Total points	Coverage
45 minutes	50	Amdahl, Gustafson, speedup, efficiency, Roofline, decomposition, collectives

NAME		STUDENT ID	
DATE		SCORE	/ 50

Instructions
- Answer every question and show essential reasoning for partial credit.
- For true/false items, an unsupported letter receives no credit.
- Unless stated otherwise, assume NVIDIA warp size 32 and ideal arithmetic.
- Keep the answer section closed until the suggested time has elapsed.

Questions

1.1 True or false with one-sentence justification — 8 points

(a) Amdahl's law keeps the problem size fixed.

(b) Efficiency can exceed 1 for any correctly measured parallel program.

(c) The Roofline ridge point is P_{peak}/B.

(d) A kernel whose operational intensity lies on the sloped Roofline region is compute-bound.

(e) Gustafson's law models scaled problem size at fixed elapsed time.

(f) More tasks always improve performance, regardless of task overhead.

(g) A barrier is a collective synchronization primitive.

(h) All-gather leaves every rank with the gathered result.

WORK AREA

1.2 Amdahl speedup, efficiency, and limit — 10 points

A program has serial fraction s=0.08 and an ideally parallel fraction 0.92.

(a) Derive the fixed-size speedup formula. (b) Compute speedup and efficiency for p=16. (c) Find the infinite-processor limit.

$$T_p/T_1 = s + (1-s)/p$$

WORK AREA

1.3 Infer the serial fraction — 6 points

A fixed-size program achieves speedup 5.0 on 8 processors. Under ideal Amdahl assumptions, estimate the serial fraction and the maximum possible speedup.

WORK AREA

1.4 Gustafson scaled speedup — 5 points

A 32-processor run spends 5% of its elapsed time in serial work. Compute Gustafson's scaled speedup and explain what is held fixed.

WORK AREA

1.5 Roofline classification — 9 points

A machine has peak compute performance 12 TFLOP/s and DRAM bandwidth 600 GB/s.

(a) Find the ridge point. (b) Bound kernels with operational intensity 4 and 30 FLOP/byte. (c) State the bottleneck of each. (d) If the first kernel's intensity doubles, give its new bound.

$$P_{\text{attainable}} \le \min(P_{\text{peak}}, B \times I)$$

WORK AREA

1.6 Measured strong scaling — 7 points

The serial run takes 80 seconds. Compute speedup and efficiency at each processor count, then identify the first point below 70% efficiency.

p	2	4	8	16
T_p (s)	42	23	14	11

WORK AREA

1.7 Decomposition and communication — 5 points

For distributed matrix-vector multiplication y=Ax with row-partitioned A, choose the needed collectives and list two design checks before implementation.

WORK AREA

ANSWER SECTION

Chapter 1: Answer Key & Worked Solutions

Grade the method as well as the final result. Numerical answers show enough intermediate work to identify where an error occurred.

Quick Answer Key

Q	Topic	Short answer
1.1	True or false with one-sentence justification	T, F, T, F, T, F, T, T
1.2	Amdahl speedup, efficiency, and limit	S(16)=7.27, E=45.45%, S(infinity)=12.5
1.3	Infer the serial fraction	s=0.0857, maximum speedup=11.67
1.4	Gustafson scaled speedup	30.45; elapsed time is fixed while problem size scales
1.5	Roofline classification	ridge=20; 2.4 TF/s memory-bound; 12 TF/s compute-bound; doubled bound=4.8 TF/s
1.6	Measured strong scaling	S={1.90,3.48,5.71,7.27}; E={95.2%,87.0%,71.4%,45.5%}; first below 70%: p=16
1.7	Decomposition and communication	Broadcast x, local matvec, Gather/Allgather y; check dependencies, granularity, balance, locality, overhead

Worked Solutions

1.1 True or false with one-sentence justification — 8 points

Answer. (a) T, (b) F, (c) T, (d) F, (e) T, (f) F, (g) T, (h) T.

- Efficiency $E=S/p$ is normally at most 1; superlinear observations require a special explanation such as cache effects.
- The sloped Roofline region is memory-bound. The horizontal roof is compute-bound.
- Task granularity must amortize scheduling and synchronization overhead.

1.2 Amdahl speedup, efficiency, and limit — 10 points

Answer. Normalize serial time to $T_1=1$. The serial part still costs s; ideal division makes the parallel part cost $(1-s)/p$.

- $T_{16}=0.08+0.92/16=0.1375$.
- $S(16)=1/0.1375=7.2727$.
- $E(16)=S/16=0.4545=45.45\%$.
- As p approaches infinity, $(1-s)/p$ approaches 0, so S approaches $1/s=12.5$.

$$S_A(p) = 1 / [s + (1-s)/p]$$

$$\lim_{p \to \infty} S_A(p) = 1/s$$

1.3 Infer the serial fraction — 6 points

Answer. Solve the Amdahl equation for s rather than substituting a known s.

- $1/5 = s + (1-s)/8 = 1/8 + 7s/8$.
- $s=(0.2-0.125)/(7/8)=0.085714$.
- Maximum speedup $=1/s=11.6667$.

$$s = (1/S - 1/p) / (1 - 1/p)$$

1.4 Gustafson scaled speedup — 5 points

Answer. The parallel run time is normalized to 1. A serial machine would spend s on serial work and p(1-s) on the scaled parallel work.

- $S_G=s+p(1-s)=p-(p-1)s$.
- $S_G=32-31(0.05)=30.45$.

$$S_G(p) = p - (p-1)s$$

1.5 Roofline classification 9 points

Answer. Use consistent units: 12 TFLOP/s=12,000 GFLOP/s and 600 GB/s times FLOP/byte produces GFLOP/s.

- Ridge I*=12,000/600=20 FLOP/byte.
- At I=4, BI=2,400 GFLOP/s=2.4 TFLOP/s, below peak: memory-bound.
- At I=30, BI=18 TFLOP/s, so the 12 TFLOP/s compute roof wins: compute-bound.
- At I=8, the bound is 600x8=4.8 TFLOP/s.

$$I^* = P_{peak}/B = 20 \text{ FLOP/byte}$$

1.6 Measured strong scaling 7 points

Answer. Compute $S=T_1/T_p$ and $E=S/p$ for every column.

p	T_p	Speedup	Efficiency
2	42	1.905	95.2%
4	23	3.478	87.0%
8	14	5.714	71.4%
16	11	7.273	45.5%

Common trap. Do not compute efficiency as T_p/T_1. Efficiency is speedup divided by processor count.

1.7 Decomposition and communication 5 points

Answer. Each rank needs the full input vector x but owns only some rows of A.

- Broadcast x from the owner, or Allgather x if its pieces begin distributed.
- Compute local output rows independently.
- Gather y to one root, or Allgather y if all ranks need the full result.
- Before coding: verify dependencies, choose task granularity, estimate communication, check balance/locality, and define the serial baseline.